SMART INDIA HACKATHON 2025

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TITLE PAGE

AR SAFETY TRAINING, REIMAGINED

AR-Based Vocational Training Simulator

for Industrial Safety in Jharkhand's Mining & Manufacturing Sector

PROBLEM STATEMENT	THEME	CATEGORY	INSTITUTE CODE
SIH26041	SMART EDUCATION	SOFTWARE	C-32887

LARP CODERS

B.Tech 2nd Year • Instrumentation & Control Engineering

TEAM MEMBERS

Anushka Goswami • Aakriti Aggarwal • Dhruv Mishra
Anish Kumar • Arnab Tiwari • Venkatesh Singhal

Name

IDEA TITLE

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Proposed Solution (Describe your Idea/Solution/Prototype)

ONE SIMULATION. EVERY DECISION MATTERS.

SEE

Phone camera + marker loads a 3D industrial scene.

AR SCENE

DECIDE

Trainee chooses what to do at interactive checkpoints.

DECISION

LEARN

Consequence + feedback + score make learning measurable.

ASSESSMENT

TRAINEE JOURNEY

OPEN APP → SELECT → SCAN MARKER → HAZARD → DECIDE → FEEDBACK → SCORE

MVP scenarios: PPE Compliance • Fire Emergency • Gas Leak & Evacuation

LARPCODDERS

Name

TECHNICAL APPROACH

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FROM CAMERA INPUT TO TRAINER INSIGHT

TRAINEE → WEB APP → AR ENGINE → CHECKPOINTS → SCORING → FIRESTORE → TRAINER

TECH STACK

AR	A-Frame + AR.js
APP	HTML / CSS / JavaScript
BACKEND	Node.js + Express
DATA	Firebase Firestore
ANALYTICS	Chart.js

AR mechanism: PHONE CAMERA → MARKER DETECTED → 3D SCENE → HAZARD → USER DECISION

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FEASIBILITY AND VIABILITY

FEASIBLE ON THE GROUND

TRACKING	Lighting / marker detection	**→ Tested markers + on-screen guidance**
PERFORMANCE	Mid-range Android limits	**→ Lightweight, low-poly 3D assets**
CONNECTIVITY	Training may happen offline	**→ Local session data + later sync**
SAFETY CONTENT	Procedures must be exact	**→ Expert / SOP validation before deployment**
LANGUAGE	Comprehension can vary	**→ Hindi + Santali localisation, verified**

WHY IT WORKS: smartphone-first • marker-based AR • modular scenarios • offline-first

IMPACT AND BENEFITS

Name

IMPACT THAT CAN BE MEASURED

Target users: mining & manufacturing trainees • trainers • safety officers

01 SAFER PRACTICE

Repeat high-risk decisions without disrupting a live workplace.

02 MEASURABLE LEARNING

Decision-level scores, feedback and completion records.

03 ACCESSIBLE TRAINING

Runs on mid-range smartphones; designed for offline use.

04 SCALABLE MODEL

Add scenarios, sites and languages without rebuilding the core engine.

EXPERIENCE → DECIDE → LEARN → CERTIFY

Name

RESEARCH AND REFERENCES

RESEARCH • VALIDATION • NEXT STEP

OFFICIAL SIH	SIH26041 problem statement and submission requirements
DGMS	Mine-safety standards, guidance and statistics
SAFETY LAW	Factories Act, 1948 / Mines Act, 1952 references in the PS
TECH	A-Frame • AR.js • Node.js / Express • Firebase Firestore • Chart.js

NEXT: expand scenarios → dashboard → localisation → certification → deployment

Safety procedures & translations require qualified human validation before real-world use.